

MASTER DEVELOPMENT SPECIFICATION

Single source of truth — 12-week build

PRIMORDIA

A First-Person VR Dinosaur-Hunting Survival RPG

Unity • Universal Render Pipeline • XR Interaction Toolkit

Target platform	Meta Quest 3 — standalone Android build (on-device, not PCVR/Link)
Experience	Fully immersive first-person VR (not passthrough / MR)
Mode	Single-player — no networking
Team	6 members — Sudharsan, Ethan, Albin, Jason, Satvik, Niranjana (~1–2 days/week, Monday sync)
Timeline	12 weeks
Dino asset	Quaternius “Animated LowPoly Dinosaurs” (CC0, 6 rigged species); built swap-ready for the paid 9-species pack
Document status	Locked scope — Core / Nice-to-have / Future tiering is binding

Contents

Right-click and choose “Update Field” to build the table of contents.

1. Purpose & Vision

This document is the single source of truth for PRIMORDIA, a first-person VR dinosaur-hunting survival RPG built in Unity for the Meta Quest 3 standalone headset. It is written to be consumed directly as the basis for the team's development TODO list, task allocation, and technical reference for the entire 12-week build. Every core system is specified with numbered TODO checklists (which map one-to-one onto task-tracker tickets), decision tables, and concrete numeric parameters. The team writes the code; this document defines what to build and in what order.

1.1 The Game in One Paragraph

You wake on an uncharted island with nothing but a small knife. The island is home to six species of dinosaur, ranging from skittish grazers to an apex T-Rex. There is no story and no dialogue — the entire game is a tight survival loop: hunt a dinosaur, harvest its loot, craft better gear from that loot, level up, and use the new gear to hunt something bigger and further from safety. A campfire is your respawn point and safe haven; behind it you build and upgrade a house. Day turns to night, night is darker and more dangerous, and you craft torches and lanterns to survive it.

1.2 The Core Loop

HUNT → LOOT → CRAFT → UPGRADE → HUNT TOUGHER

Every system in this spec exists to serve that loop. If a proposed feature does not feed hunting, looting, crafting, upgrading, or the tension between them, it is out of scope.

Loop stage	What the player does	Systems that power it
Hunt	Track, approach, and kill a dinosaur using stealth, melee, or a crafted gun.	Dino AI (§5), Combat (§4.2), VR locomotion & weapon handling (§4.1)
Loot	Harvest bulk parts, a guaranteed signature part, and a chance at rare parts.	Loot & economy (§4.4)
Craft	Spend loot + Scrap at a world-space bench to build weapons, guns, armour, lights.	Crafting (§4.3), Inventory (§4.4)
Upgrade	Gain XP, level up, upgrade the campfire radius and the house tier.	Progression (§4.5), Campfire (§4.6), House (§4.7)
Hunt tougher	Travel to harder biomes further from the campfire for better loot.	Biomes & spawning (§4.9), Day/night (§4.8)

1.3 Design Pillars

- **VR-native, not a flat game in a headset.** Interaction is diegetic: weapons live on body holsters, the crafting bench is a physical object you reach into, aiming is pointing a real controller. No floating 2D HUD bars where a world-space object will do.
- **Standalone-performant or it does not ship.** Every feature must pass one gate: does it hold target framerate on a Quest 3 running the build on-device? Baked lighting, capped particle counts, hard caps on active dinos, and aggressive despawn are non-negotiable.
- **Simple but satisfying.** The team is mostly Unity beginners with ~1–2 days a week. Every system uses the standard, free, well-documented Unity path. Depth comes from tuning and the dinosaur AI, not from exotic tech.

- **Comfort is a pass/fail requirement.** This is a mixed-reality course assignment graded in part on VR comfort. No forced camera movement, no dodge-roll, telegraphed attacks, comfort vignette on all artificial locomotion.
- **Single-player, focused scope.** No multiplayer, no story, no free-form building. The locked Core / Nice-to-have / Future tiering is the contract that keeps the build shippable.

1.4 On the Numbers in This Document

READ THIS FIRST Every balance number in this specification — weapon damage, dinosaur HP, the weapon-vs-dino power curve, crafting costs, ammo counts, XP values, level thresholds, detection ranges — is a FIRST-DRAFT STARTING POINT, not a final value. They are set to be internally consistent and to give the intended progression shape, but they exist to be playtested on-device and tuned. Treat them as the opening position for balancing, and expect to change most of them. The roadmap (§12) budgets explicit tuning passes for exactly this.

2. Scope & Non-Goals

The feature set is tiered and the tiering is binding. Build strictly top-down. Never let Nice-to-have or Future work displace Core. The 12-week roadmap (§12) schedules Core first, admits Nice-to-have only if the team is genuinely ahead, and excludes Future from the 12 weeks entirely.

2.1 CORE — Must Ship (this is the graded game)

#	Core deliverable
C1	First-person VR foundation: XR rig, smooth stick locomotion + snap-turn + comfort vignette, character controller with gravity, Y-button dash dodge, crouch (physical headset-height + button toggle).
C2	Weapon roster: 8 melee + 8 guns, each with damage / range / attack-speed or fire-rate / (guns) ammo, plus per-weapon cost + level gate, tuned as a smooth progressive power curve against the dino stat ladder.
C3	Crafted ammo (the gun sink).
C4	Simple blood: hit-spray particles + capped death decal/pool. No dismemberment.
C5	All 6 dinos, one attack each, with the full custom sense system + per-dino FSM + per-dino stat table. The dino dev's graded AI — build early and protect it.
C6	Loot economy (bulk + signature + rare parts, concrete per-dino drop table) → level-gated crafting (8+8 weapons, ammo, 3-tier armour, torch/lantern) → XP / leveling (escalating curve + level→unlock table).
C7	Campfire respawn (keep all loot, no penalty) + safe-radius + kill-count upgrade quests.
C8	House prefab-tier building (loot + level gated) with per-tier rewards.
C9	Day/night cycle + craftable light sources for night (a real light, not a vision mode).
C10	Island with 3 biomes, difficulty by distance, biome-locked spawns, one biome carrying 3 dino types; kept small but made to feel big (fog, sightline terrain, distance spawning).
C11	Audio SFX + hitscan bullet VFX wired to the gameplay sound-event system.
C12	Runs on Quest 3 standalone at target framerate; world-space VR UI.
C13	Built swap-ready: data-driven stats + swappable model prefabs for the free→paid pack change (§2.5).

2.2 NICE-TO-HAVE — Only if Core is Solid

#	Nice-to-have
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N1	Execute the paid-pack swap to 9 dinos (adds Iguanodon, Pachycephalosaurus; swaps Apatosaurus → Brachiosaurus) — cheap IF built swap-ready (\$2.5).
N2	Pteranodon flyer AI — flight is off-NavMesh and the hardest behaviour; only after the ground dinos work.
N3	Carnivore–herbivore ecosystem behaviour (dinos hunt/flee each other) — sequenced AFTER player-facing AI works.
N4	JSON save/load (cross-session persistence).
N5	House trophy displays; expanded storage tiers; meat/cooking heal; clean-kill harvest bonus; extra VFX/audio/biome polish.

2.3 FUTURE — Deferred; Do NOT Build for First Submission

Written down so the design accommodates them, but explicitly excluded from the 12 weeks. Raise these with the team only after submission.

#	Future item	Design hook to leave in place now
F1	Per-dino attack variants (charge, pounce, tail-hitbox tricks).	Build the Attack state as a swappable module (\$5.5) so variants drop in without a rewrite.
F2	Debuffs / status effects (slow, bleed, blur).	Damage events carry an (unused) status payload field.
F3	Alpha / scarred dinos (recolor + boosted stats + debuff attacks). An alpha's loot = base drops + a guaranteed rare.	Dino profile is a ScriptableObject; a variant is a second asset + material swap. Document the loot rule; do not build alphas.
F4	Sleep darts / sedation system (dart weapon, dart×dino table, Sedated FSM state, sneak-to-range).	FSM state enum reserves a Sedated slot; do not implement transitions.
F5	Weapon durability / breaking — RECOMMENDED AGAINST.	Fights the power-progression fantasy (players hoard good gear, fight with junk) and adds tracking/repair/UI work. Crafted ammo is the better sink. Note the recommendation; do not build.
F6	Swing-based (physics) melee (controller velocity drives a weapon collider).	Melee damage is routed through one hit-resolver; a velocity source can replace the button trigger later.
F7	Custom tail-swing / faint animations (require Blender authoring).	Attack module reads a clip reference; a new clip is a data change.
F8	Green-tint night-vision vision mode.	Deliberately replaced by torches/lanterns — DO NOT build.

2.4 Hard Non-Goals

These are permanently out of scope for this project, not merely deferred:

- Multiplayer or any networking.
- Narrative, quests-with-story, cutscenes, or dialogue.
- Free-form / voxel building (the house is a prefab-tier swap only).
- Hand-tracking, body/optical sensors, gloves (Touch Plus controllers only).
- Dismemberment or runtime mesh slicing (dinos are single skinned meshes).
- Night-vision vision mode (replaced by real craftable lights).
- Weapon durability / breaking (crafted ammo is the intended sink instead).
- Physical swing-velocity melee (button-press melee only this build).
- PCVR / Quest Link (standalone Android build only).
- Paid middleware, custom shaders, or advanced AI frameworks.

2.5 Asset Swap-Readiness (REQUIRED architecture)

The active build ships on the free CC0 Quaternius pack (6 species). The whole game must nonetheless be architected so that switching later to the paid polyperfect “Low Poly Animated Dinosaurs” pack (9 species, ~\$20) is a DROP-IN, not a rewrite. This is a Core constraint on how every dino system is built, even though executing the swap itself is Nice-to-have (N1).

Achieve swap-readiness with three rules, enforced from day one:

Rule	What it means concretely	Why it makes the swap a drop-in
(a) Swappable model + Animator	Each dino's mesh + Animator is a CHILD of a dino prefab (the prefab holds the AI, agent, colliders, stat reference). The visual is never the root.	Swapping a species = replace the child model + Animator; the prefab, FSM, and data are untouched.
(b) Data-driven stats, never hardcoded	Every dino stat and behaviour value lives in a DinoProfile ScriptableObject (\$5.8, \$6.5). No numbers baked into scripts.	A new species is a new data asset, not a code change; the same FSM reads it.
(c) Fixed clip-role set	Animations are referenced by role — idle / walk / run / attack / death — via named clip slots on the Animator, not by hard clip names.	Relinking a new pack's clips to the five role slots is the whole animation step.

With those in place a swap = drop in the new models, relink the five clip slots, and refit colliders/scale per species. Nothing in the AI, combat, loot, or progression code changes.

2.5.1 What the Paid Pack Adds (“add on swap”)

Change on swap	Detail	Behaviour note
+ Iguanodon	New species.	Mild defensive grazer (retaliate-only, gentle).
+ Pachycephalosaurus	New species.	Defensive rammer/charger when provoked.
+ Pteranodon (flyer)	New species; aerial.	Off-NavMesh flight — the hardest AI; build last (N2).
Apatosaurus → Brachiosaurus	Sauropod replaced.	Same role: the bulk-loot wall (high HP/XP, low damage).

SCOPE These 9-species details are documented so the architecture accommodates them. Do NOT build the extra species for the first submission — the graded game is the 6-dino free-pack build. The swap and the flyer are Nice-to-have (\$2.2).

3. Technology & Project Setup

Everything below is the standard, free, well-documented Unity path chosen deliberately for a beginner team. No paid packages, no custom render features, no third-party AI.

3.1 Engine & Package Decisions

Concern	Decision	Why (beginner-oriented)
Unity version	Unity 6 LTS (or latest 2022 LTS the lab has installed) — pick one and pin it for the whole team.	One shared version avoids project-corruption from mixed editors. LTS = stable, well-documented, matches most tutorials.
Render pipeline	Universal Render Pipeline (URP).	The only sensible pipeline for standalone Quest. Built-in is slower on mobile GPUs; HDRP will not run on Quest at all.
VR framework	XR Interaction Toolkit (XRI) 3.x + XR Plugin Management + OpenXR.	Unity's own supported VR stack. Gives locomotion providers, socket interactors (holsters), ray interactors (UI) out of the box.

AI navigation	Built-in NavMesh + NavMeshAgent.	Free, visual, no code to move an agent. Perfect for FSM-driven dinos. No behaviour-tree middleware.
Animation	Mecanim Animator controllers + the Quaternion clips.	Standard. One Animator per dino species; states map 1:1 to FSM states.
Data assets	ScriptableObjects for dino profiles, recipes, loot tables, tiers.	Designer-editable in the Inspector, no recompile to tune. Central to keeping beginners productive AND to swap-readiness (\$2.5).
Dino prefab shape	Root prefab = AI + agent + colliders + stat ref; model + Animator are swappable children.	Enforces the free→paid pack drop-in swap (\$2.5). Set this convention before authoring any dino.
Save format (N1)	Unity JsonUtility → a JSON file (Nice-to-have only).	Built-in, no library. Core build holds state in memory only.
Source control	Git + Git LFS for binary assets, with a Unity .gitignore.	LFS prevents the repo bloating on FBX/textures. Agree a branching rule Monday week 1.

BEGINNER TRAP Do NOT let members open the project on different Unity versions. The first person to open on a newer editor silently upgrades serialized files and breaks it for everyone. Pin the version in the Monday-week-1 meeting and write it in the README.

3.2 Meta Quest 3 Build Setup

Getting an empty scene building and running on-device is Phase 0 and blocks everything. Do it first, before any feature work.

- ☐ 1. Install the Android Build Support module (SDK + NDK + OpenJDK) via Unity Hub for the pinned editor version.
- ☐ 2. Switch the build target to Android; set Texture Compression to ASTC.
- ☐ 3. Install XR Plugin Management; enable the OpenXR provider for Android; add the Meta Quest feature group.
- ☐ 4. Set the OpenXR interaction profile to Oculus Touch Controller (this build uses Touch Plus, controllers only).
- ☐ 5. Set Color Space to Linear, Graphics API to Vulkan (fall back to OpenGL ES3 only if a device issue appears).
- ☐ 6. Set Minimum API Level per current Meta guidance for Quest 3; set Scripting Backend to IL2CPP, ARM64 only.
- ☐ 7. Enable Developer Mode on the Quest 3 via the Meta mobile app; authorise the dev machine over USB.
- ☒ 8. Build-and-run the default sample scene to the headset on day one and confirm it launches on-device.

GATE Phase 0 is not complete until the whole team has seen an on-device build run. This de-risks the single hardest beginner hurdle before any game code exists.

3.3 Project Folder & Package Structure

Agree this folder layout in week 1 so six people are not inventing conflicting structures. Keep everything the team authors under a single top-level _Project folder.

Folder	Contents
_Project/Scenes	Bootstrap, Island (main), and per-developer test scenes.
_Project/Scripts	Sub-folders: Player, Combat, AI, Loot, Crafting, Progression, World, UI, Data.
_Project/Data	All ScriptableObject assets: DinoProfiles, Recipes, LootTables, Tiers, Quests.
_Project/Prefabs	Dino prefabs, weapon prefabs, house tiers, campfire, VFX, UI panels.
_Project/Art	Models (Quaternion FBX), Materials, Textures, VFX, Audio.
_Project/Settings	URP asset(s), XR settings, Input Actions asset, quality levels.
ThirdParty	Imported free packages kept separate from team-authored content.

BEGINNER TRAP *One Input Actions asset for the whole project, owned by Ethan (the player-mechanics dev). Multiple competing input assets are a classic beginner merge nightmare.*

4. System Specifications

One sub-section per core system. Each has a numbered TODO checklist whose items are written to become task-tracker tickets, plus the data tables the team needs to build and tune it. Section 5 (Dinosaur Behaviour) is broken out separately and treated in the greatest depth, as it is the author's graded portion.

REMINDER All numeric values below are first-draft starting points for playtesting (see §1.4).

4.1 VR Player & Weapon Handling

First-person throughout. The player's hands ARE the controllers — there is no player-body avatar to rig, animate, or drive with IK. This is a deliberate scope saving: the only animated humanoid-style content in the game is on the dinosaurs.

4.1.1 Locomotion & Movement

Movement	Binding & behaviour	Comfort treatment
Smooth move	Left thumbstick; head-relative direction; character controller with gravity.	Comfort vignette engages while moving; tunable strength.
Snap-turn	Right thumbstick left/right; 30° or 45° discrete snaps (default 45°).	Snap (not smooth) turn is the default — far lower nausea.
Teleport (fallback)	Right thumbstick forward shows an arc; release to teleport.	Comfort option for sensitive players; not the primary.
Y-button dash dodge	Y = a quick lateral/back dash in stick direction; camera stays level.	Short comfort fade; NO rotation, NO roll (see §7).
Crouch — physical	Player physically ducks; headset height drops (free, no input).	Native and comfortable; the primary crouch.
Crouch — toggle	Left thumbstick click (accessibility); lowers the rig a fixed amount.	For players who can't/won't physically crouch.
Gravity / fall	Character controller applies gravity; player falls off ledges, lands on terrain.	No fall damage this build (comfort + simplicity).

Crouching (either method) lowers the player's effective visibility height and noise output, feeding directly into the dinosaur sense model (§5.1): a crouched player is seen at a shorter range and heard far less. This makes stealth a real, mechanically supported choice.

4.1.2 Weapon Handling — Melee & Guns

Button-press melee. Pressing the attack trigger plays a swing animation on the weapon model with a live hitbox enabled only during the swing's active frames. Tap = a normal swing; optional hold = a slower heavy swing (more damage, longer active window). This is NOT physics/velocity melee — the swing is animation-driven and the same whether the player moves the controller or not. Physical swing-velocity melee is Future (F6).

Two-handed guns. Point the controller and pull the trigger for hitscan damage (no travelling projectile): a raycast from the muzzle resolves the hit instantly. Each shot plays a muzzle flash, a cheap tracer line, an impact effect, a

recoil kick on the gun model, and consumes one crafted round. Optional second-hand grip point for a steadier two-handed pose. Out of ammo → the player falls back to melee.

4.1.3 Equip Loadout / Body-Holster Hotbar

The VR-native hotbar is a set of body-holster slots, NOT a floating 2D bar. The player carries only a small equipped set on the body and reaches to a holster to draw a weapon; the full inventory holds everything else.

Slot	Location (XRI socket interactor)	Typical contents
Right hip	Right waist	Primary melee (knife / dino weapon).
Left hip	Left waist	Primary gun.
Right shoulder	Over right shoulder	Secondary gun or heavy melee.
Left shoulder	Over left shoulder	Light source (torch/lantern) or secondary.
Armour slot	Worn (not a hand slot)	Equipped armour tier occupies its own dedicated slot.

Equip 2–4 hand-accessible items at once (start at 2 holsters, expand to 4 as an early unlock). To draw: reach to the holster and grab with the grip button; to stow: return the item near the holster and release. Which item lives in which slot is reassigned from the inventory menu, Minecraft-style — drag an inventory item onto a slot. Worn armour is equipped from the inventory and occupies the armour slot.

BEGINNER TRAP *XRI socket interactors attached to the camera rig can jitter if parented wrong. Parent holster sockets to a body-follow transform that tracks the headset's horizontal position but ignores pitch/roll, so hip sockets don't tumble when the player looks down.*

4.1.4 TODO Checklist — VR Player & Weapon Handling

- ☐ 1. Build the XR Origin rig (camera offset, controllers) using XRI; confirm head + both controllers track on-device.
- ☐ 2. Add the character controller with gravity; ground the player on the terrain.
- ☐ 3. Wire smooth locomotion (left stick) via the XRI continuous-move provider; head-relative.
- ☐ 4. Wire snap-turn (right stick) via the snap-turn provider; default 45°.
- ☐ 5. Add teleport (right-stick-forward arc) as a comfort fallback provider.
- ☐ 6. Implement the comfort vignette tunnelling provider; trigger on move AND dash.
- ☐ 7. Implement the Y-button lateral dash dodge (fixed distance/time, camera level, comfort fade).
- ☐ 8. Implement physical crouch (rig reads headset height) feeding a 'crouch factor' the sense model reads.
- ☐ 9. Add the button-toggle crouch (left-stick-click) as an accessibility alternative.
- ☐ 10. Create the two-hip + two-shoulder holster sockets on a yaw-only body-follow transform.
- ☐ 11. Implement grab-from-holster / stow-to-holster with the grip button via socket interactors.
- ☐ 12. Implement button-press melee: swing animation + hitbox live only during active frames; tap vs hold.
- ☐ 13. Implement hitscan gun firing: raycast, muzzle flash, tracer, impact FX, recoil kick, ammo decrement.
- ☐ 14. Implement out-of-ammo fallback (gun still equips but cannot fire; prompt to switch to melee).
- ☐ 15. Build the inventory-menu slot reassignment (drag item → holster slot; equip armour → armour slot).
- ☒ 16. Confirm on the Quest 3 that draw/stow feels reliable and holsters don't jitter when looking down.

4.2 Combat & Damage

One shared damage pipeline serves melee and guns so balancing and Future features (status effects, physics melee) hook in at a single point.

4.2.1 Damage Pipeline

- **Hit detection.** Melee = a trigger collider on the weapon, live only during the swing's active frames. Guns = an instantaneous raycast from the muzzle. Both produce a hit event carrying: target, damage, hit point, hit normal, and an (unused this build) status payload for Future debuffs (F2).
- **Damage resolution.** The target's health component subtracts damage. For the player, equipped armour applies a flat percentage reduction first (§4.3). For a dino, the hit routes into its FSM as a Hit/Stagger trigger and an aggro/retaliation signal (§5).
- **Death.** At ≤ 0 HP the entity enters its death state: a dino plays its death clip, disables its agent + colliders, spawns the loot drop (§4.4), grants XP (§4.5), and despawns after a short delay. The player entering 0 HP triggers campfire respawn (§4.6).

4.2.2 Player Health & Armour

Parameter	First-draft value	Notes
Base player HP	100	Flat; no HP scaling with level this build (keeps balance legible).
Regeneration	None by default	Meat/cooking heal item is Nice-to-have (N4).
Armour T1 (Hide)	−10% incoming	Flat reduction, all damage types.
Armour T2 (Bone-Plate)	−25% incoming	Flat reduction.
Armour T3 (Stego-Plate)	−40% incoming	Flat reduction; endgame survivability.
Respawn	At campfire, full HP	Keep all loot, no penalty (§4.6).

4.2.3 Blood & Bullet VFX (cheap, capped)

Simple blood only — no dismemberment (dinosaurs are single skinned meshes; runtime slicing is out of scope). Every effect is a small, capped particle burst or a decal:

Effect	Implementation	Cap / budget
Hit spray	Small blood particle burst at the hit point, oriented to the hit normal.	≤ 12 particles/burst; pooled.
Death pool/decal	One decal or tiny mesh pool under a killed dino; fades out.	1 per corpse; removed on despawn.
Muzzle flash	Brief flash sprite/light at the muzzle on fire.	Single-frame flash; light optional.
Tracer	A short line-renderer along the hitscan ray for 1–2 frames.	1 per shot; pooled.
Impact	Spark on terrain/rock, blood on a dino, at the ray hit point.	≤ 8 particles; pooled.

BEGINNER TRAP *Never Instantiate/Destroy particle effects per hit — that thrashes the GC and stutters on Quest. Pool every VFX and reuse from the pool.*

4.2.4 TODO Checklist — Combat & Damage

- ☐ 1. Create a shared Health component (current/max HP, take-damage entry point, death event).
- ☐ 2. Define the hit-event data (target, damage, point, normal, status payload placeholder).
- ☐ 3. Implement the melee hitbox: enabled only during swing active frames; one hit per swing per target.
- ☐ 4. Implement the hitscan resolver: raycast, layer mask, distance = weapon range, apply damage on hit.
- ☐ 5. Implement player armour flat-percentage reduction applied before HP subtraction.
- ☐ 6. Implement dino hit reaction: route damage into the FSM Hit/Stagger + retaliation signal.
- ☐ 7. Implement death handling: death clip, disable agent/colliders, drop loot, grant XP, timed despawn.
- ☐ 8. Build pooled VFX for hit-spray, death pool/decal, muzzle flash, tracer, impact; wire to hit events.

□ 9. Wire the audible SFX layer to the same events (gun bang, melee whoosh, dino hit/roar) — see §4.10.

⚠ 10. On-device: confirm particle counts hold framerate with several dinos dying near each other.

4.3 Weapons, Armour & Crafting

Level-gated crafting at a world-space bench. A recipe is inputs (loot + counts) → one output, with a required player level. The roster below is the progressive power curve; all numbers are first-draft (§1.4).

4.3.1 Weapon Balance Identity (the core rule)

Guns hit slightly harder and reach far, but cost crafted ammo and emit a LOUD AI sound-event (every dino nearby may hear it). **Melee** is slightly weaker but free (no ammo), silent (a tiny sound-event), and instant (no reload or fire-rate gap). **Both stay viable the entire game through this ammo + stealth trade-off. Melee must NOT out-damage guns.**

As tiers rise, guns gain fire rate and range (plus modest damage) while melee gains damage — so the late-game choice is heavy silent bursts (melee) versus rapid loud ranged fire (guns), not one strictly beating the other.

4.3.2 Melee Weapon Roster (8 weapons)

Damage = per-hit HP removed. Range is short across the board (melee reach in metres). Attack speed = swings/second (higher is faster). Eight rungs so the power curve steps gradually against the dino ladder.

Tier	Weapon	Dmg	Range (m)	Atk spd	Craft cost (first-draft)	Lvl
T1	Knife	12	1.2	2.0	Starting item (free)	1
T2	Wooden Spear	18	2.0	1.4	6 Bone, 4 Hide	3
T3	Bone Club	26	1.5	1.6	8 Bone, 4 Hide	5
T4	Claw Blade	34	1.4	1.8	4 Sickle Claw, 8 Bone	7
T5	Horn Pike	44	2.2	1.2	2 Trike Horn, 10 Bone, 6 Scrap	9
T6	Spike Maul	56	1.6	1.0	3 Tail-Spike, 4 Back-Plate, 8 Scrap	12
T7	Plated Cleaver	68	1.7	1.1	3 Back-Plate, 6 Frill Plate, 10 Scrap	14
T8	Rex-Fang Greatblade	84	1.9	1.0	1 Apex Fang, 2 Rex Skull, 12 Scrap	16

4.3.3 Gun Roster (8 weapons)

Damage = per-hit HP removed (slightly above the melee tier it sits beside). Range is long. Fire rate = shots/second. Guns consume the listed ammo. Eight rungs mirroring the melee ladder so the two lines stay balanced.

Tier	Gun	Dmg	Range (m)	Fire rate	Ammo (cap)	Craft cost (first-draft)	Lvl
T1	Scrap Revolver	22	35	1.5	Light (6)	10 Scrap, 2 Bone	4
T2	Bone Sidearm	30	40	2.2	Light (10)	14 Scrap, 4 Bone	6
T3	Pump Carbine	40	50	1.6	Light (8)	18 Scrap, 3 Claw	8
T4	Hunting Rifle	52	70	1.2	Heavy (5)	22 Scrap, 3 Horn	10
T5	Auto Carbine	44	55	5.0	Heavy (24)	26 Scrap, 4 Frill Plate	12
T6	Plated Shotgun	72	18	1.0	Shell (6)	28 Scrap, 6 Back-Plate	14
T7	Marksman Rifle	80	85	1.4	Heavy (8)	30 Scrap, 3 Tail-Spike	15
T8	Apex Rifle	98	90	2.2	Heavy (12)	34 Scrap, 1 Apex Fang, 3 Rex Skull	17

4.3.4 Weapon-Damage-vs-Dino-HP Power Curve (8 rungs)

This is the balance backbone: it shows how many hits each weapon tier needs to kill each dino, so the team can tune toward the intended feel — a gradual eight-step climb, always slightly under-gunned until the top tier. Cells show approximate hits-to-kill (dino HP from the per-dino stat table §4.5.4; melee/gun damage from the rosters above), shown as melee / gun.

Design targets: each weapon tier meaningfully answers the next dino tier with no spiky gaps. Tier 1 clears the low dinos (Para/Raptor/Trike) but CANNOT realistically beat a T-Rex; mid tiers handle the Trike/Stego band and hurt a T-Rex in a risky fight; the top tier can near-one-shot the T-Rex and grind the sauropod wall.

Weapon tier (melee dmg / gun dmg)	Para 60	Raptor 45	Trike 140	Stego 170	Apato 420	T-Rex 520
T1 (12 / 22)	5 / 3	4 / 3	12 / 7	15 / 8	35 / 20	44 / 24
T2 (18 / 30)	4 / 2	3 / 2	8 / 5	10 / 6	24 / 14	29 / 18
T3 (26 / 40)	3 / 2	2 / 2	6 / 4	7 / 5	17 / 11	20 / 13
T4 (34 / 52)	2 / 2	2 / 1	5 / 3	5 / 4	13 / 9	16 / 10
T5 (44 / 44)	2 / 2	2 / 2	4 / 4	4 / 4	10 / 10	12 / 12
T6 (56 / 72)	2 / 1	1 / 1	3 / 2	4 / 3	8 / 6	10 / 8
T7 (68 / 80)	1 / 1	1 / 1	3 / 2	3 / 3	7 / 6	8 / 7
T8 (84 / 98)	1 / 1	1 / 1	2 / 2	3 / 2	5 / 5	7 / 6

TUNING These hits-to-kill are the dials the balance passes turn. With 8 rungs the steps should feel gradual, not spiky — if any single tier jump feels too large, nudge that weapon's damage or the neighbouring dino's HP. Keep the shape: under-gunned until T8, and the sauropod always a slow chip-kill.

4.3.5 Armour (3 tiers, flat reduction)

Tier	Name	Reduction	Craft cost (first-draft)	Lvl
T1	Hide Armour	−10%	8 Para Hide, 6 Bone	3
T2	Bone-Plate Armour	−25%	3 Frill Plate, 12 Bone, 6 Scrap	8
T3	Stego-Plate Armour	−40%	4 Back-Plate, 4 Tail-Spike, 10 Scrap	12

Apatosaurus loot is NOT armour — it is bulk building material for the house (§4.4, §4.7).

4.3.6 Light Sources (2 tiers) & Crafted Ammo

Item	Effect	Craft cost (first-draft)	Lvl
Torch / Lantern (T1)	A warm point light on the off-hand/weapon; modest range.	4 Sickle Claw, 3 Bone, 2 Scrap	5
Bright Lantern / Headlamp (T2)	A brighter, wider point light; longer range for night travel.	1 Rex Skull, 8 Scrap	14
Light Ammo (×20)	Feeds T1/T2 sidearms.	6 Scrap	4
Heavy Ammo (×10)	Feeds rifles / carbine / apex rifle.	8 Scrap, 1 Bone	10
Shells (×8)	Feeds the Plated Shotgun.	8 Scrap, 1 Back-Plate	15

Light sources are a real Unity light on the player/weapon — NOT a night-vision vision mode (that is explicitly cut, F8). Ammo is the deliberate gun sink; there is no weapon durability (F5).

4.3.7 Crafting UI & Flow (world-space VR)

- Crafting happens at a physical bench object near the campfire; the player opens it by pointing the controller ray at it (or a nearby touch).
- The panel is a world-space canvas at a comfortable distance and height, showing recipe cards: output icon, input list with owned/required counts, required level, and a Craft button that is disabled (greyed) until level + materials are met.
- Recipes below the player's level are visible but locked (shown with the gating level) so progression goals are legible.
- Crafting consumes inputs from the inventory and adds the output; a short confirm SFX/VFX plays. No timers — crafting is instant to keep the loop tight.

4.3.8 TODO Checklist — Weapons, Armour & Crafting

- ☐ 1. Create the Recipe ScriptableObject (inputs+counts, output, required level, category).
- ☐ 2. Author all 8 melee, 8 gun, armour, light, and ammo recipes as data assets from the tables above.
- ☐ 3. Create the Weapon data asset (damage, range, attack-speed/fire-rate, ammo type+capacity, tier).
- ☐ 4. Build weapon prefabs for each melee + gun (Jason's 2D gun art → 3D by Ethan).
- ☐ 5. Implement the armour equip + flat-reduction hook into the damage pipeline (§4.2).
- ☐ 6. Implement the two light-source tiers as pooled Unity lights on the off-hand/weapon.
- ☐ 7. Implement crafted-ammo items and the inventory counts that guns decrement.
- ☐ 8. Build the world-space crafting panel: recipe cards, owned/required counts, level lock, Craft button.
- ☐ 9. Wire crafting to consume inputs + grant output + play confirm feedback.
-  10. Run a first balance pass on-device against the 8-rung power-curve table; log every number you change.

4.4 Loot & Economy

Three resource classes, and every drop feeds at least one of three sinks. Tougher dinos drop more and rarer loot; easy grazers drop little and common.

4.4.1 Resource Classes

- **Bulk.** Hide and Bone (Meat optional, N4) from every dino, scaling with body size/tier. Apatosaurus is the main bulk Bone/Hide source and the backbone of house-building.
- **Species-signature parts.** One guaranteed unique drop per species, used for that species' themed gear, house plates, and campfire kill-quests.
- **Rare / trophy.** A low-chance premium part, chance scaling with tier. The T-Rex's Rex Skull is the marquee rare.
- **Scrap / Metal (world-gathered).** NOT from dinos — scavenged from crates / scrap points scattered across the biomes. Feeds all guns and all ammo, so guns aren't pure dino parts.

4.4.2 Signature-Part Table (6 active)

Species	Signature part(s)	Primary use
T-Rex	Apex Fang / Rex Skull	Endgame weapon + gun, T2 light, house top tier.
Triceratops	Horn / Frill Plate	Horn Pike, Bone-Plate armour.
Stegosaurus	Back-Plate / Tail-Spike	Spike Maul, Stego-Plate armour, shells.
Apatosaurus (sauropod)	Giant Bone / Thick Hide	Bulk build materials (house).
Velociraptor	Sickle Claw / Pelt	Claw Blade, T1 light.
Parasaurolophus	Hide / Crest	Hide armour, early bulk.

On the paid-pack swap (N1), add: Iguanodon → Thumb Spike / Hide; Pachycephalosaurus → Skull Dome / Bone; Pteranodon → Beak / Wing Membrane; and Brachiosaurus replaces Apatosaurus as the bulk source. These are documented for swap-readiness, not built now.

4.4.3 Per-Dino Drop Table (first-draft)

Each kill rolls: the guaranteed signature part(s), a weighted bulk roll, and a tier-scaled rare chance. Quantities and rare odds rise with difficulty tier.

Dino (tier)	Guaranteed signature	Bulk roll (Hide/Bone)	Rare chance	Rare part
Parasaurolophus (Easy)	1 Crest	1–2 Hide	3%	Pristine Crest
Velociraptor (Easy–Med)	1 Sickle Claw	1 Hide	8%	Alpha Pelt
Triceratops (Med)	1 Horn	2–3 Bone	6%	Frill Plate (extra)
Stegosaurus (Med)	1 Back-Plate	2–4 Bone, 1 Hide	10%	Twin Tail-Spike
Apatosaurus (Med wall)	2 Giant Bone	6–10 Bone, 4–6 Hide	10%	Ancient Bone
T-Rex (Apex)	1 Apex Fang	3–5 Bone, 2–3 Hide	20%	Rex Skull

The sauropod's oversized bulk roll is deliberate — it is the main house-building material source, the payoff for a slow chip-kill. Alpha-variant drops (Future, F3) would be base drops + a guaranteed rare — documented here, but alphas are not built this submission.

4.4.4 The Three Sinks

Sink	Consumes	Detail
Crafting	Signature parts + bulk + Scrap	Weapons, guns, ammo, 3-tier armour, torches (§4.3).
House tiers	Apato bulk mats + signature plates	Prefab-tier upgrades (§4.7).
Campfire kill-quests	Specific signature parts	Grow the safe radius (§4.6).

4.4.5 Inventory & Storage

- Stacking model: identical items stack with a per-stack cap (e.g. 99 for common bulk, lower for rare parts); the inventory is a simple slot grid.
- Base inventory capacity is modest; total storage grows with house tier (§4.7).
- World-space grid UI opened via controller ray; drag to move/assign, matching the holster reassignment flow (§4.1).

4.4.6 TODO Checklist — Loot & Economy

- ☐ 1. Niranjana: create Item data assets for every bulk part, signature part, rare part, Scrap, and ammo type.
- ☐ 2. Create a LootTable ScriptableObject per dino (guaranteed + weighted bulk + rare chance).
- ☐ 3. Implement the drop roll on dino death; spawn a pooled pickup or auto-add to inventory.
- ☐ 4. Place world Scrap/scavenge points across biomes; implement gather-on-interact.
- ☐ 5. Build the inventory data model (slots, stacks, caps) and the world-space grid UI.
- ☐ 6. Link storage capacity to house tier.
-  7. Playtest drop rates: is early loot scarce enough to make hunting meaningful, not grindy?

4.5 Progression / Leveling

XP comes only from kills, scaled by dino tier. The level curve escalates — each level costs more than the last — and each level unlocks higher crafting and build tiers. All values first-draft (§1.4).

4.5.1 XP-per-Dino (first-draft)

Dino	Tier	XP reward
Parasaurolophus	Easy	20
Velociraptor	Easy–Med	35 (each; packs add up)
Triceratops	Medium	45
Stegosaurus	Medium	70
Apatosaurus (sauropod)	Medium (wall)	150 (enormous HP → high XP payout)
T-Rex	Apex	300

Note the two toppers reward differently: the T-Rex pays high XP for a lethal fight; the sauropod pays high XP for a long, low-risk chip-kill — you earn it through patience and ammo/durability of approach, not danger.

4.5.2 Escalating Level Curve (first-draft)

XP to go from level N to N+1 follows a simple escalating rule the team can tune with one multiplier: $\text{cost}(N \rightarrow N+1) = \text{round}(80 \times 1.35^{(N-1)})$. Concrete thresholds:

Level reached	2	3	4	5	6	8	10	13	15	17
XP for this level	80	108	146	197	266	484	882	2160	3937	7175
Cumulative XP	80	188	334	531	797	1815	3800	9100	15700	28500

TUNING The 1.35 growth factor and 80 base are the two dials. Raise the factor to stretch the grind, lower it to speed leveling. Validate against how many kills of each tier a level actually costs on-device.

4.5.3 Level → Unlock Table (first-draft)

Level	Unlocks
1	Knife (start); 2 holster slots.
3	Wooden Spear; Hide Armour (T1); House Lv1.
4	Scrap Revolver; Light Ammo; 4th holster slot.
5	Bone Club; Torch/Lantern (T1 light).
6	Bone Sidearm; Campfire Tier 2.
7	Claw Blade.
8	Pump Carbine; Bone-Plate Armour (T2); House Lv2.
9	Horn Pike.
10	Hunting Rifle; Heavy Ammo; Campfire Tier 3.
12	Spike Maul; Auto Carbine; Stego-Plate Armour (T3).
13	House Lv3.
14	Plated Cleaver; Plated Shotgun; Shells; Bright Lantern/Headlamp (T2 light).
15	Marksman Rifle.
16	Rex-Fang Greatblade.
17	Apex Rifle (endgame).

4.5.4 Per-Dino Stat Table (first-draft)

The combat/progression shape of the six dinos in one place (senses live in the AI master table, §5.8). Peers cluster with slight variation, and the two toppers sit above the ladder for DIFFERENT reasons — read the design-intent column.

Dino	HP	Damage	Move / Chase (m/s)	XP	Tier	Design intent
Parasaurolophus	60	0 (flees)	1.5 / 6.0	20	Easy	Skittish starter; never fights.
Velociraptor	45	18	3.0 / 6.5	35	Easy–Med	Low HP, fast, dangerous in packs.
Triceratops	140	28	1.2 / 4.5	45	Med	Retaliate-only; hits a touch HARDER than Stego.
Stegosaurus	170	26	1.0 / 3.5	70	Med	Retaliate-only; a touch TANKIER than Trike.
Apatosaurus (sauropod)	420	22	0.9 / 3.0	150	Med (wall)	Enormous HP + high XP, LOW damage — a passive wall you chip for bulk loot.
T-Rex	520	45	2.2 / 5.5	300	Apex	High HP + HIGH damage — the lethal apex; the campfire exception.

SHAPE Trike/Stego are a deliberately close pair (Stego +30 HP tankier, Trike +2 damage punchier). The sauropod and T-Rex both top the ladder, but the sauropod is a low-danger HP/XP wall while the T-Rex is a genuine lethal threat — keep that contrast through tuning.

4.5.5 TODO Checklist — Progression

- ☐ 1. Add XP reward to each dino profile; grant XP on death (§4.2).
- ☐ 2. Implement the level curve (base + growth factor) and current-XP / current-level tracking.
- ☐ 3. Implement the unlock gate: recipes check required level; unlocked recipes become craftable.
- ☐ 4. Show a simple diegetic level/XP readout (e.g. on the wrist or the crafting panel).
-  5. Playtest pacing: does each biome's difficulty line up with the level its gear unlocks at?

4.6 Campfire (Respawn + Safe Zone)

The campfire is home base: respawn point, safe zone, and the anchor for the house and crafting bench. Respawn keeps ALL loot — there is no death penalty.

4.6.1 Respawn & Safe-Radius Logic

- **Respawn.** On player death, fade out, teleport the rig to the campfire spawn point, restore full HP, keep the entire inventory and all equipped gear. No loot dropped, no XP lost.
- **Safe radius.** A cylinder around the campfire inside which small/medium dinos will not spawn and will not path in; if one is mid-chase and the player crosses in, it breaks off at the boundary. The radius grows with campfire tier.
- **Apex exception.** The T-Rex ignores the safe radius — it can enter and attack at the campfire. This keeps apex encounters tense and prevents the campfire trivialising the endgame.

4.6.2 Kill-Count Upgrade Quests & Campfire Tiers

The campfire upgrades via kill-count quests of the form “kill X of species Y”, consuming specific signature parts on completion. Each tier grows the radius. Per-tier asset requirements are listed so Albin (environment) knows what prefab to source/build.

Tier	Radius	Upgrade requirement (first-draft)	Asset requirement (env dev)
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1 (start)	6 m	— (built at start)	Small fire pit; low warm point light + ember particles (capped).
2	10 m	Kill 8 Parasaurolophus + 5 Triceratops; give 3 Crest.	Larger stone ring; brighter light; small log seating.
3	16 m	Kill 6 Velociraptor + 3 Stegosaurus; give 2 Back-Plate.	Big bonfire; wide light radius; banner/marker posts for the boundary.

4.6.3 TODO Checklist — Campfire

- ☐ 1. Place the campfire + spawn point; make it the initial respawn location.
- ☐ 2. Implement respawn (fade, teleport, full HP, keep all loot).
- ☐ 3. Implement the safe-radius trigger: suppress small/medium spawns, break off chases at the boundary.
- ☐ 4. Implement the T-Rex apex exception (ignores the radius).
- ☐ 5. Create Quest data assets (target species, count, signature-part cost, radius reward).
- ☐ 6. Implement kill-count tracking + quest turn-in that grows the radius and swaps the campfire tier prefab.
- ⚠️ 7. Confirm on-device the boundary break-off reads clearly to the player (no dinos stuck at the edge).

4.7 Base / House Building

NOT free-form building — a prefab-tier swap gated by loot + level. Upgrading swaps the whole house prefab for a bigger one and grants that tier's rewards. The spec defines requirements only; sourcing/building the actual prefabs is Albin's job (free kits exist), and the spec provides no 3D models.

Tier	Lvl	Cost (first-draft)	Asset requirement (footprint)	Protection & rewards
Lv1 Shelter	3	20 Giant Bone, 15 Thick Hide	Small hut prefab; single room.	Full night shelter from small dinos; basic storage (+1 row); respawn point moves here if built.
Lv2 Cabin	8	40 Giant Bone, 25 Hide, 4 Frill Plate	Medium cabin; walls + door.	Larger safe radius bonus; expanded storage (+3 rows); 10% crafting-cost discount.
Lv3 Lodge	13	60 Giant Bone, 6 Back-Plate, 1 Rex Skull	Large lodge; multi-room.	Faster respawn; big storage (+6 rows); 20% crafting discount; unlocks trophy displays (N3).

Apatosaurus bulk loot (Giant Bone / Thick Hide) is the primary house currency — the reason a huge passive grazer is worth hunting.

4.7.1 TODO Checklist — House

- ☐ 1. Albin: source/build the three house-tier prefabs to the footprints above.
- ☐ 2. Create a HouseTier data asset (level gate, cost, storage bonus, radius bonus, discount, rewards).
- ☐ 3. Implement the build/upgrade interaction near the campfire (consume cost, swap prefab).
- ☐ 4. Apply per-tier rewards (storage rows, safe-radius bonus, crafting discount, respawn speed).
- ☐ 5. Wire night shelter: inside the house at night, small-dino threat is suppressed (apex excepted).
- ⚠️ 6. Confirm each prefab swap keeps the player/campfire alignment and doesn't clip terrain.

4.8 Day/Night Cycle

A looping day/night cycle that makes night meaningfully harder without expensive rendering. Night is navigable ONLY via craftable lights + campfire glow — explicitly NOT a night-vision vision mode (F8).

Parameter	First-draft value	Notes
Full cycle length	10 min (7 day / 3 night)	Long enough to hunt, short enough to see night in a demo.

Lighting method	Rotate one directional 'sun' + swap ambient/skybox tint	Cheap; drives brightness. Bake static shadows, no realtime shadows.
Night visibility	Lower ambient + fog tint	Darker, shorter sight lines; NO post-processing night-vision.
Light sources	Craftable torch/lantern + campfire glow	The intended way to see at night (§4.3).
Night danger	Spawn density +25%, aggression bump	More dinos, more aggressive (§5).
Safety	Campfire/house suppress small dinos at night	Apex (T-Rex) still ignores it.

4.8.1 TODO Checklist — Day/Night

- ☐ 1. Implement the cycle clock (configurable length) driving a normalized time-of-day value.
- ☐ 2. Rotate the directional sun + lerp ambient/skybox tint from the time value; keep shadows baked.
- ☐ 3. Lower night ambient + add a light fog tint for reduced night visibility (no post-processing).
- ☐ 4. Feed time-of-day into the sense model (night sight reduction) and spawner (density/aggression bump).
- ⚠️ 5. On-device: confirm night is atmospheric but framerate holds with lights active.

4.9 Map / Biomes & Spawning

One island, three biomes radiating outward from the campfire; difficulty scales with distance. Species are biome-locked, with Biome 2 carrying three dino types. The playable area is kept deliberately small — cheap to build and to run — but engineered to FEEL large (§4.9.2). Visuals stay cheap: terrain + a few tree/rock prefab types per biome, no dense high-poly foliage.

Biome	Difficulty	Species (biome-locked)	Character
1 — Forest & Clearings	Easy (spawn)	Parasaurolophus (flees), Triceratops (retaliate-only)	Holds the campfire + starting area; green forest with open grassy clearings.
2 — Open Plains / Wetlands	Medium (3 types)	Apatosaurus (passive bulk), Stegosaurus (defensive), Velociraptor packs (sound-driven)	Open sightlines; the bulk-farming and armour-mat biome.
3 — Dark Jungle / Volcanic Badlands	Hard / Apex	T-Rex (sight-driven, long aggro), heavier Velociraptor packs	Dark, enclosed, dangerous; endgame loot. Volcano backdrop + small lava set-piece.

4.9.1 Volcano Backdrop & Lava Set-Piece (near-zero cost)

- The volcano is a distant skybox/horizon backdrop plus ONE small walkable lava set-piece — atmosphere for almost no performance cost. It is NOT a real rendered volcanic region with simulation.
- The lava set-piece can be a simple emissive plane + a hazard trigger (light damage on contact) if the team wants interactivity; otherwise purely scenic.
- This sells 'dangerous badlands' at the horizon while the actual playable Biome 3 stays small and cheap.

4.9.2 Making a Small Map Feel Big

Technique	How	Bonus
Fog + limited draw distance	URP fog + a modest camera far-plane so you never see the whole island at once.	Also makes night scarier and cuts render cost.
Sightline-blocking terrain	Curve the terrain, use hills/tree walls so each biome is revealed as you travel, not all upfront.	Creates a sense of discovery and hides the map edges.

Distance-based spawning	Dinos appear on approach and despawn when far (§4.9.3), so a small area feels populated without all species active at once.	Directly serves the active-dino cap (§7).
Volcano horizon	The distant volcano implies a much larger world beyond the playable bounds.	Free scale (§4.9.1).

4.9.3 Spawning & Performance Rules

Rule	First-draft value	Reason
Per-biome spawn cap	Biome1 4, Biome2 5, Biome3 4	Bounds active agents per area.
Global active-dino cap	8 simultaneously	Hard Quest 3 performance ceiling (§7).
Spawn distance	Spawn when player within ~40–60 m of a spawn node	Don't populate the far island.
Despawn distance	Despawn when > ~80 m and out of sight	Frees agents; pools the prefab.
Campfire suppression	No small/med spawns inside the safe radius	Apex excepted (§4.6).
Night modifier	+25% density within caps, aggression bump	Night danger (§4.8).

BEGINNER TRAP Bake the NavMesh AFTER the terrain and biome props are placed, and keep slopes gentle. A steep, unbaked, or hole-ridden island is the #1 cause of dinos refusing to move or sliding through geometry (see Risk Register, §12).

4.9.4 TODO Checklist — Biomes & Spawning

- ☐ 1. Albin: sculpt the small island terrain with the three biome zones by distance; curve terrain to block sightlines.
- ☐ 2. Add a few cheap tree/rock prefab types per biome; keep triangle + draw-call budgets (§7).
- ☐ 3. Add the volcano skybox/horizon backdrop + one small walkable lava set-piece (optional hazard trigger).
- ☐ 4. Set up URP fog + a modest camera far-plane so the whole island is never visible at once.
- ☐ 5. Bake the NavMesh after props are placed; verify agents path across all three biomes.
- ☐ 6. Create spawn nodes tagged by biome + species; implement the biome-locked spawner with caps (Biome 2 = 3 types).
- ☐ 7. Implement distance-based spawn/despawn with prefab pooling.
- ☐ 8. Wire campfire suppression + apex exception + night density/aggression modifier into the spawner.
- ☒ 9. On-device: patrol each biome and confirm the global active-dino cap is never exceeded.

4.10 Audio & Effects (SFX + VFX)

Two layers that are wired together but built by different people, so keep them conceptually separate:

(a) The gameplay sound-event — an INVISIBLE loudness value the dino AI 'hears' (§5.1). Owned by San (the AI author). It has no sound; it is data.

(b) The audible sound effect — the actual audio the player hears through the headset. Nirranjan's audio task. It has no gameplay effect on AI.

The same action fires both. Keep them in lockstep but do not conflate them: a gun both makes a loud bang (audible) and emits a loud sound-event (AI); a silenced-feeling melee makes a quiet whoosh and a tiny sound-event.

4.10.1 Action → (Audible SFX / AI Sound-Event) Map

Action	Audible SFX	AI sound-event loudness
Gunshot	Loud bang + echo	Loud (large radius) — alerts most nearby dinos.
Melee swing/hit	Quiet whoosh / thud	Tiny (small radius) — near-stealth.

Footstep — sprint	Loud footfalls	Medium radius.
Footstep — walk	Soft footfalls	Small radius.
Footstep — crouched	Near-silent	Near-zero radius (stealth).
Crafting	Craft clink/confirm	Small radius.
Dino roar / call	Species SFX	(Audible only; ambience + threat cue.)

Footstep loudness scales with movement mode (sprint > walk > crouch ≈ silent), tying stealth gameplay to both what the player hears and what dinos hear.

4.10.2 TODO Checklist — Audio & Effects

- ☐ 1. San: define the sound-event API (position, loudness/radius) that dinos subscribe to (§5.1).
- ☐ 2. Emit sound-events from gun fire, melee, footsteps (by mode), crafting — with the loudness map above.
- ☐ 3. Niranjan: gather/assign cheap SFX clips for each action + dino calls; wire to the same triggers.
- ☐ 4. Implement footstep loudness scaling with sprint/walk/crouch for BOTH layers.
- ☐ 5. Confirm the hitscan bullet VFX (muzzle/tracer/impact, §4.2) fire alongside the gun SFX.
- ⚠️ 6. On-device: verify audio doesn't cause frame hitches and 3D spatialisation reads correctly.

5. Dinosaur Behaviour (Priority Section)

AUTHOR'S GRADED PORTION This is Sudharsan's (San's) individually graded AI system and the deepest part of the specification. It is built entirely on the standard free Unity stack: NavMesh + NavMeshAgent for movement, one finite-state machine (FSM) per dinosaur, and a custom two-channel sense model. No behaviour-tree or utility-AI middleware — an FSM is easier for this team to build, cheaper on Quest, and far easier to explain and defend in the report.

USE SAN'S OWN FSM Ignore any bundled Wander/AI script that ships with an asset pack (free or paid). The AI **MUST** be San's own FSM — it is the graded work. Asset packs contribute models and clips only.

Everything here is built data-driven and model-agnostic (§2.5): the FSM reads a DinoProfile ScriptableObject and drives whatever model+Animator child the prefab holds, so the same AI runs the free 6 dinos now and the paid 9 later with no rewrite.

5.1 The Sense Model

Each dino perceives the player through two independent, cheap channels: sight and hearing. Either channel can raise alertness; together they drive the FSM. Both are budgeted to run at a low tick rate (§5.9), not every frame.

5.1.1 Sight — FOV Cone + Line-of-Sight

Sight is a two-test gate. A player is 'seen' only if BOTH pass:

Test	What it checks	Notes
1. FOV cone	Is the player within the species' sight range AND inside its horizontal FOV angle (half-angle each side of facing)?	Cheap: distance check + a dot-product against the forward vector.
2. Line-of-sight ray	A single raycast from the dino's eye to the player; blocked by terrain/props = not seen.	Only cast if test 1 passes — saves raycasts.

Sight is modulated by conditions, applied as multipliers on the effective sight range:

Condition	Effect on sight range	Rationale
Night	× 0.5	Dinos see the player at half range in the dark (§4.8).
Player crouching	× 0.5	Lower profile; harder to see (§4.1).
Player behind cover	Blocked by the LoS ray	Handled naturally by test 2 — no extra logic.
Player standing, day	× 1.0	Baseline.

Multipliers stack (crouching at night → ×0.25 range), rewarding stealth play strongly at night.

5.1.2 Hearing — Sound-Event Radius

Hearing is event-driven, not polled. Player actions publish a sound-event at a world position with a loudness radius (the gameplay sound-event from §4.10 — invisible data, not audio). Each dino compares its distance to the event against its own hearing radius scaled by the event loudness; if within, it hears it and investigates.

Player action	Event loudness radius (first-draft)	Stealth consequence
Gunshot	40 m	Loud — the price of ranged power (§4.3).
Sprinting	18 m	Fast but noisy.
Walking	9 m	Moderate.
Melee swing/hit	5 m	Near-silent — the stealth weapon.
Crafting	6 m	Do it inside the safe radius.
Crouched movement	2 m	Effectively silent.

A dino 'hears' an event when: $\text{distance}(\text{dino}, \text{event}) \leq \text{dinoHearingRadius} \times (\text{eventRadius} / \text{referenceRadius})$. Louder events reach dinos that a quiet event would not. This is the mechanical backbone of noise discipline.

5.2 Per-Species Behaviour Profiles

One profile per dino (6), each a ScriptableObject so the author tunes values in the Inspector without recompiling. Design intent per species:

- **T-Rex — sight-dominant apex.** Very long sight range, wide-ish FOV; detects and chases from afar; high HP AND high damage — the lethal threat; the campfire/house-safety EXCEPTION (§4.6); slow to give up.
- **Velociraptor — hearing-dominant pack hunter.** Large hearing radius, modest sight; low HP but fast; reacts strongly to sound; hunts in packs and coordinates (§5.5).
- **Triceratops — defensive, retaliate-only.** Ignores the player until attacked, then turns and defends with its single gore attack; hits a touch HARDER than Stego; disengages when the player leaves.
- **Stegosaurus — defensive, retaliate-only.** As Triceratops but a touch TANKIER and slower; armoured bulk that punishes melee range with its tail-side attack.
- **Apatosaurus — huge passive grazer (the wall).** Enormous HP and high XP but LOW damage; extremely slow to react; retaliates sluggishly if attacked but mostly just grazes. A low-danger bulk-loot piñata you chip down over time.
- **Parasaurolophus — skittish grazer.** Flees the moment it detects the player by either channel; never fights; fast burst away then resumes grazing at distance.

5.2.1 Per-Species Sense & Behaviour Table (first-draft)

The at-a-glance profile. Sight range in metres; FOV is the full cone angle; hearing in metres; speeds in m/s. HP, damage, XP, and difficulty tie to §4.

Species	Sight r	FOV	Hear r	Aggression	Move	Chase	Atk r	Dmg	HP	XP	Tier
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Parasaurolophus	22	160°	16	Flee	1.5	6.0	—	0	60	20	Easy
Velociraptor	18	140°	28	Aggressive (pack)	3.0	6.5	2.0	18	45	35	Easy–Med
Triceratops	16	120°	10	Retaliate	1.2	4.5	3.0	28	140	45	Med
Stegosaurus	14	110°	9	Retaliate	1.0	3.5	3.2	26	170	70	Med
Apatosaurus	12	100°	8	Passive (wall)	0.9	3.0	4.0	22	420	150	Med
T-Rex	40	150°	22	Apex hunter	2.2	5.5	3.5	45	520	300	Apex

TUNING This table and the master table (§5.8) are the author's primary balancing surface. Every value here is first-draft and expected to move under on-device playtesting (§1.4).

5.3 The Finite-State Machine (per dino)

Every dino runs the same eight-state FSM; species differ by which transitions are enabled and by their profile values. Each state maps to a NavMesh action and one Quaternius animation clip (idle / walk / run / attack / death — there is exactly one attack clip and no faint/tail clip, so design around that).

State	NavMesh action	Animation	Enter when	Exit when
Idle / Graze	Stationary (or tiny local moves)	idle	Spawn; nothing sensed	Senses player → Alerted; timer → Patrol
Patrol / Wander	Roam between local waypoints	walk	Idle timer elapses	Senses player → Alerted; arrives → Idle
Alerted / Investigate	Move toward last sense position	walk	Sight or hearing trips	Confirms player → Chase/Flee; loses → Patrol
Chase	Set destination = player, repath	run	Aggressive/apex confirms player; retaliator was hit	In attack range → Attack; loses player/timeout → Patrol
Attack	Stop; face player; run Attack module	attack	Player within attack range	Player leaves range → Chase; attack ends → re-eval
Flee	Set destination away from player	run	Skittish senses player; (optional) low-HP panic	Safe distance + timer → Patrol
Hit / Stagger	Brief stop	(flinch via attack/short clip)	Takes damage	Short timer → Chase (retaliate) or Flee
Dead	Disable agent + colliders	death	HP ≤ 0	After despawn delay (drop loot, grant XP)

5.3.1 Per-Species Transition Enablement

Species	Idle/Graze	Patrol	Alerted	Chase	Attack	Flee	Hit
Parasaurolophus	✓	✓	✓	—	—	✓	✓ (→Flee)
Triceratops	✓	✓	on hit	✓ (retaliate)	✓	—	✓ (→Chase)
Velociraptor	✓	✓	✓	✓	✓	opt. low-HP	✓ (→Chase)
Stegosaurus	✓	✓	on hit	✓ (retaliate)	✓	—	✓ (→Chase)
Apatosaurus	✓	✓	on hit (slow)	✓ (slow)	✓	—	✓ (→Chase, slow)
T-Rex	✓	✓	✓	✓ (long)	✓	—	✓ (→Chase)

Retaliators (Trike, Stego, Apato) leave Idle/Patrol only via a hit — their Alerted entry is gated on damage, not on sensing. Skittish Para never enters Chase/Attack. This is data-driven: the FSM code is identical; a per-species flag set enables transitions.

5.4 The Attack State — a Swappable Module

Build the Attack action as a separate, swappable module, NOT hardcoded into the FSM. The FSM's Attack state calls a referenced 'attack behaviour' object; this build ships exactly one — a melee 'basic attack' (play the single attack clip, enable a damage hitbox on the clip's active frames, apply the profile's damage if the player is still in range). Because the FSM only knows 'run the attached attack', Future variants (charge, pounce, tail-hitbox — F1) drop in as new attack modules with zero FSM rewrite.

- The attack must be clearly telegraphed: a visible wind-up at the start of the clip so a player using physical room-scale dodging (or the Y-dash) can react. Attacks must not be too fast — this is a VR comfort requirement (§7), not just balance.
- Damage applies on the clip's active frames only, via a hitbox enabled/disabled by animation events (or a timed window if events are fiddly for the team).
- One attack clip per species means all six share this one module shape; per-species differences are just damage, range, and clip timing from the profile.

5.5 Player-Behaviour Reactions

Reaction	Rule	Which species
Noise discipline	Crouch → barely detectable; walk → moderate; sprint/gunfire → loud (§5.1.2). Detection is a direct function of the player's movement/attack mode.	All (hearing-weighted most for Velociraptor).
Retaliation / aggro escalation	A hit forces Alerted→Chase on retaliators and refreshes aggro on hunters, extending pursuit.	Trike, Stego, Apato (trigger); Raptor, T-Rex (escalate).
Pursuit of a fleeing player	Hunters keep chasing while they retain sight/recent sound, up to a give-up timer, then return to Patrol. T-Rex has the longest timer.	Velociraptor (medium), T-Rex (long).
Raptor grouping / coordination	Raptors share alerts within a group radius: one detecting the player alerts packmates, who converge; simple staggered approach so they don't stack on one point.	Velociraptor only.
Night aggression	At night, detection ranges/aggression get the night bump and predators give up more slowly (§4.8).	Velociraptor, T-Rex most.
Skittish flee	On any detection, Para bursts away to a safe distance, then resumes grazing; re-flees if re-detected.	Parasaurolophus.

5.5.1 Raptor Pack Coordination (detail)

- Group raptors at spawn into a pack sharing one lightweight 'pack blackboard' (last known player position + alert flag).
- When any packmate senses the player, it writes the position and raises the alert; all packmates within the group radius transition to Alerted/Chase toward it.
- To avoid all raptors piling onto one point, give each a small angular offset around the player so they approach from a spread — cheap, and reads as 'coordination' without any real pathfinding cleverness.
- Keep pack size small (2–3) to respect the active-dino cap (§5.9, §7).

5.6 On-Swap Species Profiles (Nice-to-have, documented for swap-readiness)

SCOPE These three profiles are authored ONLY if the paid-pack swap (N1) is executed after Core is solid. They are documented now so the FSM/sense architecture already accommodates them — each is just a new DinoProfile asset + a model child (§2.5), no FSM rewrite. The flyer is the single hardest AI and is built last (N2).

Species	Role / aggression	Behaviour notes	Build risk
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Iguanodon	Mild defensive grazer	Retaliate-only and gentle — reuses the Trike/Stego defensive pattern with softer stats. Trivial to add.	Low
Pachycephalosaurus	Defensive rammer/charger	Retaliate-only, but its 'attack' is a charge — a good first use of the swappable Attack module (a charge variant, F1-style) without touching the FSM.	Low–Med
Pteranodon (flyer)	Aerial predator/scavenger	OFF-NavMesh flight — does not use NavMeshAgent ground pathing. Needs a separate simple flight mover (waypoints/steering at a fixed altitude, swoop to attack). Reuses the SAME sense model + FSM states; only the movement backend differs.	HIGH — build last

Flyer approach (when built): keep it a lightweight steering mover between air waypoints with a swoop-down for the Attack state; it senses via the same sight/hearing channels and runs the same FSM (Idle=circle, Chase=approach, Attack=swoop, Flee=climb away). Do NOT try to make NavMesh handle flight.

5.7 Carnivore–Herbivore Ecosystem (Nice-to-have — sequence AFTER player AI)

SEQUENCING Build this ONLY once all six dinos react correctly to the PLAYER. It is strong report material and extends the same FSM + sense system, but it must never delay the graded player-facing AI. Keep it in a clearly separated, optional layer that can be disabled.

The ecosystem layer lets dinos perceive EACH OTHER using the same sense channels, adding two optional states that reuse existing movement:

Added behaviour	Who	How it reuses the existing system
Hunt-prey	Carnivores (T-Rex, Velociraptor)	Sense system flags a herbivore as prey; Chase/Attack retarget from player to the herbivore.
Flee-from-predator	Herbivores (Para, Trike, Stego, Apato)	Same Flee/retaliate logic, triggered by sensing a carnivore instead of the player.

- Target selection: a carnivore picks the nearest sensed valid target (player or herbivore); keep the arbitration simple and readable.
- Cap the extra cost: ecosystem sensing shares the same low tick rate and the same active-dino cap — it must not add per-frame work.
- It should be possible to toggle the whole layer off for a stable player-only demo build.

5.8 Master Tuning Table (all AI parameters in one place)

Every per-species numeric knob collected for balancing. This is the single sheet the author tunes from; all values first-draft (§1.4). Times in seconds; ranges in metres.

Parameter	Para	Trike	Raptor	Stego	Apato	T-Rex
Sight range (m)	22	16	18	14	12	40
FOV (full angle)	160°	120°	140°	110°	100°	150°
Hearing radius (m)	16	10	28	9	8	22
Move speed (m/s)	1.5	1.2	3.0	1.0	0.9	2.2
Chase speed (m/s)	6.0	4.5	6.5	3.5	3.0	5.5
Attack range (m)	—	3.0	2.0	3.2	4.0	3.5
Attack damage	0	28	18	26	22	45
Attack wind-up (s)	—	0.6	0.35	0.7	0.8	0.7
Attack cooldown (s)	—	1.8	1.0	2.0	2.2	1.6

Health (HP)	60	140	45	170	420	520
XP reward	20	45	35	70	150	300
Give-up timer (s)	—	4	8	4	5	14
Flee trigger	on sense	—	opt low-HP	—	—	—
Respects campfire	n/a	yes	yes	yes	yes	NO (apex)

5.9 AI Performance Notes (Quest 3 standalone)

Concern	Rule	Why
Sense tick rate	Run sight/hearing checks ~4–10× per second per dino, staggered across frames — NOT every frame.	Raycasting every dino every frame will blow the frame budget.
Stagger	Distribute dinos across tick 'buckets' so they don't all sense on the same frame.	Avoids periodic frame spikes.
Active-dino cap	Hard cap of 8 active NavMeshAgents at once (§7, §4.9).	The single biggest AI performance lever.
LOD / despawn	Distant dinos reduce animation/sense rate; beyond despawn distance they pool out (§4.9).	Keeps far agents nearly free.
Repath throttle	Recompute Chase destination a few times per second, not every frame.	SetDestination every frame is wasteful.
Animator culling	Set Animators to cull when off-screen.	Skips animation work for unseen dinos.

BEGINNER TRAP Do not give every dino its own Update() raycast loop. Use one AI manager that ticks a slice of the dino list each frame (time-slicing). This one decision is the difference between a smooth demo and an unshippable one.

5.10 TODO Checklist — Dinosaur Behaviour

- ☐ 1. Create the DinoProfile ScriptableObject holding every parameter in the master table (§5.8).
- ☐ 2. Author all six profile assets with the first-draft values; keep the dino prefab model+Animator as a swappable child (§2.5).
- ☐ 3. Implement the sight channel: distance + FOV dot-product gate, then a single LoS raycast.
- ☐ 4. Apply sight multipliers for night and player crouch; confirm cover blocks via the LoS ray.
- ☐ 5. Implement the hearing channel subscribing to the sound-event API (§4.10) with per-species radius.
- ☐ 6. Build the eight-state FSM as shared code driven by a per-species transition-enable flag set (ignore any bundled Wander script).
- ☐ 7. Map each state to its NavMesh action + the Animator's role clip slot (idle/walk/run/attack/death).
- ☐ 8. Implement the swappable Attack module (single melee attack: telegraph, active-frame hitbox, damage).
- ☐ 9. Implement retaliation (hit → Alerted/Chase on retaliators; aggro refresh on hunters).
- ☐ 10. Implement Flee (skittish Para; optional low-HP panic) and the give-up timers.
- ☐ 11. Implement raptor pack coordination (shared blackboard, converge with angular spread, size 2–3).
- ☐ 12. Build the AI manager that time-slices sense/repath ticks across dinos; enforce the active-dino cap.
- ☐ 13. Add Animator off-screen culling and distance-based LOD/despawn hooks.
- ☐ 14. (Nice-to-have, after Core) On paid-pack swap: author *Iguanodon* + *Pachycephalosaurus* profiles (data + model child, no FSM change).
- ☐ 15. (Nice-to-have, build LAST) *Pteranodon* flyer: a separate off-NavMesh flight mover reusing the same sense model + FSM.
- ☐ 16. (Nice-to-have, after player AI) Add the carnivore–herbivore ecosystem layer, toggleable.

⚠ 17. On-device: verify each species senses, alerts, chases/flees, attacks, staggers, and dies correctly at framerate.

6. Data / Save-State Schema

The Core build holds all state in memory for a single session. JSON save/load (N2) serialises this same model to disk — design the model now so the save is a later add-on, not a refactor. Tables below give field, type, and description.

6.1 Player State

Field	Type	Description
currentHP	float	Current health; reset to max on respawn.
maxHP	float	Base 100 (§4.2).
level	int	Current level (§4.5).
currentXP	int	XP toward the next level.
equippedArmourTier	int (0–3)	0 = none; drives damage reduction.
holsterSlots	int (2–4)	Unlocked hand-accessible slots.
holsterContents	itemId[]	Item in each holster slot.
position / respawnPoint	Vector3	Current transform / campfire respawn.

6.2 Inventory & Items

Field	Type	Description
itemId	string / enum	Stable id (e.g. bone, apex_fang, heavy_ammo).
itemClass	enum	Bulk / Signature / Rare / Scrap / Ammo / Weapon / Armour / Light.
stackCount	int	Quantity in this stack; capped per item.
inventorySlots	slot[]	Grid of {itemId, stackCount}.
storageCapacityRows	int	Grows with house tier (§4.7).

6.3 Weapons, Armour, Ammo (data assets)

Field	Type	Description
weaponId	string	Stable id.
weaponType	enum	Melee / Gun.
damage	float	Per-hit HP removed.
range	float	Melee reach or hitscan distance (m).
attackSpeed / fireRate	float	Swings or shots per second.
ammoType	enum (guns)	Light / Heavy / Shell.
ammoCapacity	int (guns)	Magazine size.
tier / requiredLevel	int	Power tier and unlock level.
armourTier / reductionPct	int / float	Armour only: 1–3 and its flat reduction.

6.4 Recipes

Field	Type	Description
outputId / outputQty	string / int	What crafting produces.
inputs	{itemId, count}[]	Required materials and counts.
requiredLevel	int	Level gate.
category	enum	Weapon / Gun / Ammo / Armour / Light.

6.5 Dino Profiles (mirrors the master tuning table)

Field	Type	Description
speciesId	enum	One of the six species.
sightRange / fovAngle	float	Sight cone (§5.1).
hearingRadius	float	Base hearing radius (§5.1).
moveSpeed / chaseSpeed	float	NavMesh agent speeds.
attackRange / attackDamage	float	Attack module inputs (§5.4).
attackWindup / attackCooldown	float	Telegraph + rate (§5.8).
health / xpReward	float / int	Combat + progression.
aggressionType	enum	Flee / Retaliate / Passive / Aggressive / Apex.
giveUpTimer	float	Pursuit persistence.
respectsCampfire	bool	False only for T-Rex.
transitionFlags	flags	Which FSM transitions are enabled (§5.3.1).
lootTableRef	ref	This species' drop table (§4.4).

6.6 Campfire, House, Quests

Field	Type	Description
campfireTier / safeRadius	int / float	Current tier and its radius (§4.6).
houseTier	int (0–3)	Current house tier (§4.7).
activeQuests	quest[]	{targetSpecies, requiredCount, currentCount, signatureCost, radiusReward}.
killCounts	dict<species,int>	Running kills per species for quests.
timeOfDay	float (0–1)	Day/night clock (§4.8).

6.7 JSON Save Format (Nice-to-have, N2)

One save object aggregating the above, written with Unity JsonUtility to a single file in persistentDataPath. Save on key events (level-up, tier change) and on quit; load on launch. No third-party library.

- Root: { player, inventory, unlockedRecipes, campfire, house, quests, timeOfDay, saveVersion }.
- Include a saveVersion int so later format changes can migrate gracefully.
- Data assets (weapons, dino profiles, recipes) stay in the project — the save stores only ids + runtime state, never the asset definitions.

7. VR Comfort & Performance Spec

Consolidated from the requirements threaded through every system. These are pass/fail gates, not aspirations: a feature that fails either the comfort rules or the performance budget must be simplified until it passes.

7.1 Comfort Rules (course-graded)

- **Snap-turn by default.** Smooth turn is off by default; snap-turn (45°) is the primary. Offer smooth turn only as an explicit opt-in.
- **No forced camera movement, ever.** The camera moves only from the player's own head or deliberate stick input. No cutscene pans, no knockback that rotates the view.
- **Dash, never roll.** The Y-dash keeps the camera level with a short comfort fade. No dodge-roll, no tumbling — a roll is a severe nausea trigger and would fail the comfort bar.
- **Comfort vignette on all artificial motion.** Tunnelling vignette on smooth locomotion AND the dash; strength tunable, on by default.
- **Telegraphed, not-too-fast attacks.** Every dino attack has a visible wind-up (§5.4) so players can physically dodge; attack speeds are capped for reaction time.
- **Comfortable world-space UI.** Panels sit at a readable distance and height, never pinned to the face; interacted via controller ray.
- **Physical crouch supported.** Real ducking works (and is rewarded by stealth); a toggle exists for players who can't physically crouch.

7.2 Quest 3 Standalone Performance Budget

Budget item	Target (first-draft)	Notes
Frame rate	72 Hz locked (revisit 90 Hz only if headroom)	Dropping frames in VR causes nausea — framerate is non-negotiable.
Lighting	Baked lightmaps; NO realtime shadows	Realtime shadows are a top mobile-GPU cost.
Draw calls	~150–250 per frame	Use static/dynamic batching + shared materials.
Triangles on screen	~300–500 K	Low-poly Quaternius dinos + cheap terrain help.
Active dinos	Hard cap 8 (§5.8)	The single biggest perf lever.
Textures	≤ 1–2 K, ASTC compressed	Atlas where possible.
Particles	Capped per effect, pooled (§4.2)	No per-hit Instantiate/Destroy.
Culling	Occlusion + distance spawn/despawn (§4.9)	Don't render/simulate the far island.
Post-processing	Minimal / none	No expensive full-screen effects; night is done with lighting, not post.

THE GATE For every feature ask: (1) Does it run on Quest 3 standalone at 72 Hz? (2) Can a mostly-beginner team build it at ~1–2 days/week across 12 weeks? If either answer is no, simplify the feature and record the decision.

8. UX / Visual & Interaction Design

8.1 Controller Button Map (Meta Quest 3 Touch Plus)

One canonical binding table owned by Ethan (single Input Actions asset, §3.3). Left/right may be swapped for left-handed players as an option.

Control	Binding	Action
Left thumbstick	Push	Smooth move (head-relative).
Left thumbstick	Click	Crouch toggle (accessibility).
Right thumbstick	Left / Right	Snap-turn 45°.
Right thumbstick	Forward	Teleport arc (comfort fallback).
Y button	Press	Dash dodge (lateral/back, camera level).
X button	Press	Open inventory / crafting (world-space panel).
A button	Press	Interact / loot / confirm.
B button	Press	Reload (guns) / cancel.
Right trigger	Pull	Primary use: melee swing or gun fire (held item).
Left trigger	Pull	Off-hand use (e.g. hold light / secondary).
Grip (either)	Hold	Grab from / stow to holster (§4.1).
Right trigger	Hold (melee)	Heavy swing (optional).

BEGINNER TRAP Bind actions through the Input Actions asset / XRI action references, not hard-coded device buttons. It keeps left-handed remapping and future rebinding trivial and avoids scattering device-specific code.

8.2 VR UI Approach

- All menus (inventory, crafting, campfire quests, house upgrade) are world-space canvases interacted with via the XRI ray interactor (point + trigger) or direct touch when close.
- Menus appear anchored in front of the relevant object (crafting at the bench, upgrades at the campfire/house) rather than as a full-screen overlay, preserving immersion and comfort.
- Keep panels large-text and low-density; VR text must be legible at distance.

8.3 Diegetic HUD Readouts

- Health: a diegetic readout on a wrist/watch or the armour, not a screen-space bar.
- Ammo: shown on the gun model (a small counter) so it's visible while aiming.
- Level/XP: on the wrist readout or surfaced in the crafting panel.
- Quest/kill-count progress: on the campfire panel, checked when the player visits.

8.4 Cheap-for-Standalone Art Notes

- Lean on the low-poly Quaternius dinos; recolour via material swaps for variety (and for Future alpha variants) rather than new meshes.
- A few tree/rock prefab types per biome, instanced and batched; no dense high-poly foliage.
- Unlit or simple lit URP materials; bake lighting; avoid transparency overdraw.
- Reuse one particle atlas for blood/impact/muzzle; reuse one audio set across similar actions.

8.5 Making the Small Map Feel Big (recap)

The playable island is deliberately small (cheap to build and to run) but engineered to feel large. The four techniques, consolidated from §4.9.2 as a design/UX concern:

- **Fog + limited draw distance** — you never see the whole island at once; also makes night scarier and cuts render cost.
- **Sightline-blocking terrain** — curved terrain and hill/tree walls reveal each biome as you travel, creating discovery and hiding the map edges.
- **Distance-based spawning** — dinos appear on approach and despawn when far, so a small area feels populated without all species active at once.
- **Volcano horizon backdrop** — a distant volcano implies a much larger world beyond the playable bounds, for near-zero cost.

9. 12-Week Phased Roadmap

Milestone-based and paced for six people at ~1–2 days/week with a Monday sync before the practical. Core lands first: the XR rig and an on-device build come before any gameplay, then the hunt→loot→craft→level loop, then the dino AI is built early and protected. Nice-to-have is attempted only after Core is stable; Future is excluded from the 12 weeks. Explicit balance-tuning passes are budgeted — the weapon/dino/XP numbers need on-device playtesting.

Phase	Weeks	Milestone	Scope	Primary owner(s)
Phase 0	Week 1	Foundation & on-device build	Pin Unity version; set up Git+LFS, folder structure, URP, XRI, and the swap-ready dino-prefab convention (\$2.5). Get the empty sample scene building and RUNNING on the Quest 3 for everyone. Import the Quaternius pack. First Monday (Satvik runs it): agree conventions.	Whole team; Ethan leads setup
Phase 1	Weeks 2–3	VR player foundation	XR rig, smooth move + snap-turn + teleport fallback, comfort vignette, character controller + gravity, Y-dash, physical + toggle crouch, holster grab/stow. Deliverable: comfortable movement + draw/stow a placeholder item on-device.	Ethan (pair a teammate on the rig); San starts sense model
Phase 2	Weeks 3–5	Dino AI core (protected priority)	Sense model (sight + hearing), the eight-state FSM, NavMesh movement, swappable Attack module, per-species profiles + stat table for all 6, raptor packs, AI manager time-slicing + active-dino cap — all data-driven/model-agnostic. Deliverable: all 6 dinos sense/chase/flee/attack/die on-device at framerate.	San (lead); Niranjan on profile data entry
Phase 3	Weeks 4–6	Combat, weapons & first loot	Damage pipeline, hitscan guns + button melee, armour reduction, capped blood + bullet VFX, the first few weapon tiers, drop-on-death + basic inventory. Deliverable: kill a dino, get loot, take damage.	Ethan + Jason (guns/VFX); Albin starts the island
Phase 4	Weeks 6–8	Crafting, progression & full 8+8 roster	World-space crafting, full 8 melee + 8 gun + armour + light + ammo roster + recipes, XP + escalating level curve + unlock gates. Deliverable: the full hunt→loot→craft→level loop end-to-end.	Niranjan (data/UI) + Ethan; Jason weapon stats; San tunes dinos
Phase 5	Weeks 7–9	World: biomes, campfire, house, day/night	Small-but-feels-big three-biome island (fog, sightline terrain, volcano backdrop + lava set-piece) + biome-locked spawner with caps + spawn/despawn, campfire respawn + safe-radius + kill-quests + tiers, house prefab tiers, day/night + lights. Deliverable: the full world loop.	Albin (lead); Niranjan on quest/UI data
Phase 6	Weeks 9–10	Audio/VFX pass & integration	Niranjan wires the audible SFX layer to San's sound-event system, footstep loudness by mode, dino calls; confirm all VFX pooled + capped. Deliverable: a cohesive audio-visual build.	Niranjan (audio) + San (event API); Jason VFX polish
Phase 7	Weeks 10–11	Balance tuning passes (on-device)	Dedicated playtesting: does the 8-rung curve feel gradual? Can T1 gear NOT beat a T-Rex and top gear near-one-shot it? Is the sauropod a slow chip-kill? Tune weapon/dino/XP/cost numbers against the tables in \$4–\$5. Log every change.	Niranjan leads QA; San + Niranjan tune data; whole team plays

Phase 8	Weeks 11–12	Comfort/perf hardening, report & PPT	On-device comfort + framerate profiling to budget (§7), fix perf hotspots, finalise the report and presentation. Nice-to-have attempted ONLY if Core is solid, in order: the paid-pack swap (9 dinos) → Pteranodon flyer → ecosystem → JSON save. Deliverable: submission build + docs.	Satvik (report/PPT/video); whole team; San if swap/flyer attempted
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PROTECT THE AI The dino AI (Phase 2) is scheduled early on purpose — it is the graded portion and the riskiest system. Do not let world-building or polish slip it later.

10. Team Task Allocation

Six named members mapped to workstreams. Sudharsan's (San's) dinosaur-AI tasks are given the most detail as the graded portion; Ethan, the only experienced dev, deliberately shares knowledge to a beginner to avoid a bottleneck (§12). Every member — including Niranjana — has load-bearing work the project depends on.

Member	Workstream	Phase-by-phase tasks
Sudharsan ("San")	ALL dinosaur AI + (N) ecosystem & flyer	P1 start sense model; P2 sense (sight+hearing), 8-state FSM, NavMesh movement, swappable Attack module, all 6 profiles + stat table, raptor packs, AI manager time-slicing + active-dino cap — built data-driven/model-agnostic (§2.5); P4/P7 tune the master table on-device; P6 define the sound-event API for Niranjana; (N, after Core) the paid-pack profiles, the Pteranodon flyer (off-NavMesh, built last), and the carnivore–herbivore layer. Owns §5 end-to-end — the graded work.
Ethan	Lead / VR player, weapons, input, glue	P0 lead setup & on-device build + swap-ready prefab convention; P1 rig/locomotion/dash/crouch/holsters (pair a beginner on the rig); P3 combat + guns/melee + turning Jason's 2D guns into 3D; P4 crafting hookup + core-systems glue. Single point of failure — documents the rig/input setup and keeps systems modular.
Albin	Map / environment & world prefabs	P3 start the island; P5 the small-but-feels-big three-biome terrain (fog, sightline curves), the volcano backdrop + lava set-piece, biome-locked spawner (caps, spawn/despawn), NavMesh bake, and sourcing/placing campfire/house/light/nature prefabs per tier.
Jason	Weapons dev / 2D artist	P3–P4 design the 8 guns (and dino-weapon concepts) in 2D for Ethan to build as 3D; owns weapon stat data and the muzzle/tracer/impact VFX hookup; P6 VFX polish; supports UI icons.
Satvik	Documentation & presentation lead	Throughout: keep this spec + the TODO board current and run the Monday syncs; capture demo footage; P8 write the report and build the presentation/video. Coordinates playtest notes in P7.
Niranjana	Gameplay content, audio & QA	P2 dino profile data entry; P4 loot/recipe/XP/stat ScriptableObject data + world-space UI panels (inventory/crafting/quests); P6 the audio pass — source/record and wire gun, melee, and footstep sounds into San's sound-event system; P7 lead playtesting + balance passes. A real, load-bearing role.

11. Testing Strategy

Testing is on-device first — in-editor VR simulation hides the performance and comfort issues that actually matter on Quest 3.

Test area	What & how	Pass criteria
On-device Quest 3	Build-and-run to the headset every phase; never trust editor-only.	Feature works on-device, not just in the editor.

Framerate / perf profiling	OVR Metrics / Unity profiler on-device; worst case = many dinos + night + VFX.	Holds 72 Hz within the \$7 budget.
VR comfort	Multiple testers of varying sensitivity; check turn, dash, attacks, UI distance.	No forced motion; no nausea reports on default settings.
AI behaviour	Per species: does it sense, alert, chase/flee, attack, stagger, die correctly? Raptor packs converge? Retaliators only aggro on hit? T-Rex ignores campfire?	Each species matches its \$5 profile.
Balance playtesting	Play the curve: T1 gear vs easy dinos vs T-Rex; top gear vs T-Rex; leveling pace vs biome difficulty.	T1 can't beat T-Rex; top gear near-one-shots it; progression feels 'slightly under-gunned until endgame'.
Economy / loot	Drop rates, sink balance (crafting vs house vs quests), inventory/storage.	Loot feels meaningful, not grindy; every drop has a use.
Save/load (if N2 built)	Save, quit, relaunch, verify state restored.	Player/inventory/tiers/quests restore correctly.

12. Risk Register

Risk	Impact	Mitigation
Standalone performance blowout	High — nausea + fails the perf gate	Hard active-dino cap (8), baked lighting, pooled/capped VFX, AI time-slicing, distance despawn; profile on-device every phase (\$7).
VR motion sickness	High — fails the comfort bar	Snap-turn default, vignette on all artificial motion, dash-not-roll, telegraphed attacks, comfortable UI; test with sensitive users early (\$7, \$11).
Scope creep on limited part-time hours	High — Core doesn't ship	Binding Core/Nice/Future tiering (\$2); roadmap front-loads Core; Nice-to-have only if ahead; Future explicitly excluded. Enforce at each Monday sync.
Balance takes longer than expected	Medium — curve feels flat/unfair	Budget dedicated tuning passes (Phase 7); all numbers first-draft + centralised in tables (\$4–\$5) so tuning is data-only, not code changes.
NavMesh on an uneven / volcanic island	Medium — dinos stuck / sliding	Keep slopes gentle; bake AFTER props placed; keep the lava set-piece off the NavMesh or as a simple hazard trigger; verify pathing across all biomes; adjust agent step/slope settings (\$4.9).
Ethan is a single experienced-dev bottleneck	High — blocks the team if unavailable	Pair a beginner with Ethan on the XR rig from Phase 0; document the rig/input setup in the README; keep systems modular so San, Albin, and Niranjan can progress independently.
Swap-readiness not maintained	Medium — the paid-pack swap becomes a rewrite	Enforce the three \$2.5 rules from day one (swappable model child, data-driven stats, fixed clip-role set); review at a Monday sync before the roster grows.
Beginner Unity mistakes (versions, input, merges)	Medium — lost time	Pin the editor version; one Input Actions asset (Ethan owns it); Git LFS + a branching rule agreed week 1; call out traps inline (this doc).

13. Appendix — Unity APIs & Packages

The concrete free tools this project relies on. All are first-party Unity or free packages — no paid middleware.

Tool / API	Used for	Note
XR Interaction Toolkit (XRI)	Rig, controller input, locomotion, socket (holster) & ray (UI) interactors	The supported VR stack; the backbone of \$4.1 and \$8.

XR Plugin Management + OpenXR	Enabling the Quest 3 / Meta feature set for Android	Configure in Project Settings (§3.2).
NavMesh + NavMeshAgent	Dino movement and pathfinding	Free, visual; drives every FSM state's movement (§5).
Universal Render Pipeline (URP)	The render pipeline for standalone VR performance	Required for Quest; enables the perf budget (§7).
Animator / animation clips	Playing the Quaternius idle/walk/run/attack/death clips	One Animator per species, states mapped to the FSM (§5.3).
ScriptableObject	Dino profiles, recipes, loot tables, weapon/armour/tier data	Designer-editable, no recompile — central to tuning (§4–§6).
Input System (Input Actions)	One canonical binding asset for all controls	Single asset owned by Ethan (§3.3, §8.1).
Particle System	Capped, pooled blood / muzzle / tracer / impact VFX	No per-hit Instantiate/Destroy (§4.2).
JsonUtility (built-in)	The Nice-to-have JSON save/load (N2)	No third-party library; serialises the §6 model.
Git + Git LFS	Source control with binary asset handling	Agree branching + LFS tracking in week 1 (§3.1).
OVR Metrics / Unity Profiler	On-device framerate & performance profiling	For the pass/fail perf gate (§7, §11).

End of specification. This document is the single source of truth for the PRIMORDIA 12-week build; keep it updated as decisions change at the Monday syncs.